CS402/MA492 Exam preparation

The exam paper will consist of approx. 16–20 problems divided among four questions; you should answer all four questions. The following is a list of problems in the *style* of exam questions; actual questions may differ. All topics covered in the lectures may appear on the exam, even if they do not feature on this list.

This file was last modified on 2019/04/01.

Problems

1. The ciphertext string

   `N_YM_OVRMDODIB`

   was produced using a shift cipher over the 27-letter alphabet

   `ABCDEFGHIJKLMNOPQRSTUVWXYZ_`

   We know that the plaintext symbol `K` corresponds to the ciphertext symbol `F`. Find the plaintext string.

2. A 2-dimensional Hill cipher over the 27-letter alphabet

   `ABCDEFGHIJKLMNOPQRSTUVWXYZ_`

   (identified with $\mathbb{Z}_{27}$ as in the lectures) is used to encrypt the plaintext string `SURE` using the key

   $$\begin{bmatrix} 1 & 8 \\ 11 & 9 \end{bmatrix}.$$

   Find the ciphertext string.

3. An affine cipher over the 27-letter alphabet $\mathbb{Z}_{27}$

   `ABCDEFGHIJKLMNOPQRSTUVWXYZ_`

   was used to encrypt a plaintext message using the key $(2, 3)$. The resulting ciphertext string is

   `RTJJLC`

   Find the plaintext string.

4. Identify $\mathbb{Z}_{27} = \{0, \dots, 25, 26\}$ with $\{\mathtt{A}, \dots, \mathtt{Z}, \llcorner\!\lrcorner\}$. An affine cipher over $\mathbb{Z}_{27}$ with unknown key (a, b) encrypts D as W and L as F. Determine (a, b).

5. Define what is meant by a *modular inverse* of an integer. Is 3 invertible modulo 1234?

6. Is

$$\begin{bmatrix} 23 & 42 \\ 45 & 27 \end{bmatrix}$$

invertible over $\mathbb{Z}_{68}$? If so, find its inverse over $\mathbb{Z}_{68}$.

7. Explain in which sense Hill ciphers and Vigenère ciphers are both special cases of (possibly higher-dimensional) affine ciphers.

8. Explain what is mean by *letter frequency analysis* and give an example of a cryptosystem vulnerable to it.

9. Describe what is meant by a *brute force attack* and describe a scenario where this is feasible.

10. Bob says that a general substitution cipher over an n-letter alphabet Σ is far more secure than a shift cipher since the key space is much larger.
    a) What are the sizes of the key spaces of a shift cipher and substitution cipher on Σ, respectively?
    b) Do you agree with Bob? Explain your answer.

11. Formally define what is meant by a (symmetric) *cryptosystem.*

12. State *Kerckhoff's principle.*

13. Explain what is meant by a *side-channel attack.*

14. Define what it means for a cryptosystem to be *perfectly secure.*

15. State (without proof) *Shannon's Theorem* on perfectly secure cryptosystems.

16. Consider a (symmetric) cryptosystem with non-empty finite sets $\mathcal{P}$, $\mathcal{C}$, and $\mathcal{K}$ of plaintexts, ciphertexts, and keys, respectively and with encryption functions $e_k\colon \mathcal{P} \to \mathcal{C}$ and decryption functions $d_k\colon \mathcal{C} \to \mathcal{P}$ indexed by $k \in \mathcal{K}$. Show that $|\mathcal{C}| \geq |\mathcal{P}|$.

17. Describe what is meant by a *one-time pad.* In your answer, describe one advantage and one disadvantage of using one-time pads.

18. Briefly describe *block ciphers* and *stream ciphers.* Provide one example of each type of cipher.

19. Explain what is meant by a *linear feedback shift register* of length L. Explain how LFSRs can be used as part of a stream cipher.

20. Define what is meant by the *connection polynomial* of a linear feedback shift register.

21. Define what is meant by a *non-singular* LFSR.

22. Explain what is meant by a *primitive* polynomial over $\mathbb{Z}_2$ and how these are related to LFSRs.

23. Explain how a known-plaintext attack on an LFSR-based stream cipher can sometimes be used to recover the connection polynomial.

24. A linear feedback shift register of length 3 with connection polynomial

$$C(X) = 1 + X^2 + X^3$$

over $\mathbb{Z}_2$ is used to produce a pseudo-random sequence of binary digits with initial values $s_0 = s_1 = s_2 = 1$. Determine the least period of the resulting infinite sequence $s_0, s_1, \ldots$.

25. *"For symmetric cryptosystems, key distribution scales badly as the number of parties grows."* Explain.

26. Formally define what is meant by a *public-key cryptosystem*.

27. Describe the key generation step, the encryption step, and the decryption step of the *RSA* cryptosystem.

28. Alice encrypts the message $m = 2$ using the RSA cryptosystem and Bob's public key $(n, e) = (65, 14)$. Find the corresponding ciphertext as an element of $\mathbb{Z}_{65}$.

29. *"If there is an efficient algorithm for integer factorisation, then RSA is insecure."* Explain.

30. Define *Euler's totient function*

31. State (without proof) *Euler's totient theorem.*

32. State *Fermat's little theorem* and show how it folllows from Euler's totient theorem.

33. Describe the *Fermat primality test.*

34. Define what is meant by a *group.*

35. Define what it means for a group to be *abelian* (= *commutative*).

36. Define what is means for a group to be *cyclic.* Give an example of a cyclic group and an example of a non-cyclic group.

37. For which $n \geq 1$ is the group $\mathbb{Z}_n^\times$ cyclic?

38. Decribe the method of *fast exponentiation* (also called *modular exponentiation*).

39. Describe the *Diffie-Hellman key exchange* protocol.

40. Alice and Bob use the cyclic group $\mathbb{Z}_{23}^\times$ and its generator $g = 5$ to perform a *Diffie-Hellman key exchange.* Alice secretly chooses $a = 3$ and Bob secretly chooses $b = 4$. Determine their shared key (as an element of $\mathbb{Z}_{23}^\times$).

41. Describe the key generation step, the encryption step, and the decryption step of the *ElGamal* cryptosystem.

42. State Hasse's theorem on the number of points on an elliptic curve over $\mathbb{Z}_p$.

43. Let E be the set of points of an elliptic curve over $\mathbb{Q}$ defined by the equation

$$y^2 = x^3 + ax + b.$$

Describe the usual operation which turns the set of points of E into an abelian group.